

[Assignment-1]

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Section : 01

Course : CSE260
code :

Faculty : [FIC]
Initial :

Answer to the question no 1

a) $(11011.101)_2$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= (27.625)_{10}$$

b) $(0.0001)_2$

$$= 0 \times 2^{-1} + 0 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$$

$$= (0.0625)_{10}$$

Answer to the question no 2

a) $(29)_{12}$

$$= 2 \times 12^1 + 9 \times 12^0$$

$$= (33)_{10}$$

$$= (45)_7$$

$$\begin{array}{r|l} 7 & 33 \\ \hline 7 & 4 \quad \text{---} \quad 5 \\ & 0 \quad \text{---} \quad 4 \end{array} \quad \uparrow$$

$$b) (10110111)_5$$

$$= (1 \times 5^7) + (0 \times 5^6) + (1 \times 5^5) + (1 \times 5^4) + (0 \times 5^3) + (1 \times 5^2) + (1 \times 5^1) + (1 \times 5^0)$$

$$= (81906)_{10}$$

$$= (10\ 333\ 33\ 02)_4$$

$$\begin{array}{r|l} 4 & 81906 \\ \hline 4 & 20476 \quad \text{---} 2 \\ \hline 4 & 5119 \quad \text{---} 0 \\ \hline 4 & 1279 \quad \text{---} 3 \\ \hline 4 & 319 \quad \text{---} 3 \\ \hline 4 & 79 \quad \text{---} 3 \\ \hline 4 & 19 \quad \text{---} 3 \\ \hline 4 & 4 \quad \text{---} 3 \\ \hline 4 & 1 \quad \text{---} 0 \\ \hline & 0 \quad \text{---} 1 \end{array}$$

Answer to the question no 3

$$\begin{array}{r} (412)_9 \\ + (134)_9 \\ \hline (546)_9 \end{array}$$

$$\begin{array}{r} \overset{3}{\cancel{4}} \overset{9+}{\cancel{1}} \overset{9+}{2} \\ (134)_9 \\ \hline (267)_9 \end{array}$$

$$\begin{array}{r} (412)_9 \\ \times (134)_9 \\ \hline \overset{1}{1} \overset{1}{3} \overset{1}{4} 8 \\ 1336 \times \\ 412 \times \times \\ \hline (56418)_9 \end{array}$$

$$\begin{array}{r} 1 \\ 9 \overline{) 16} \\ \underline{9} \\ 7 \end{array}$$

$$\begin{array}{r} 1 \\ 9 \overline{) 13} \\ \underline{9} \\ 4 \end{array}$$

$$\begin{array}{r} 1 \\ 9 \overline{) 12} \\ \underline{9} \\ 3 \end{array}$$

$$\begin{array}{r} 1 \\ 9 \overline{) 10} \\ \underline{9} \\ 1 \end{array}$$

$$(412)_9$$

$$= 4 \times 9^2 + 1 \times 9^1 + 2 \times 9^0$$

$$= (335)_{10}$$

$$(134)_9$$

$$= 1 \times 9^2 + 3 \times 9^1 + 4 \times 9^0$$

$$= (112)_{10}$$

$$\begin{array}{r} (335)_{10} \\ + (112)_{10} \\ \hline (447)_{10} \end{array}$$

$$\begin{array}{r} (335)_{10} \\ - (112)_{10} \\ \hline (223)_{10} \end{array}$$

$$\begin{array}{r} (335)_{10} \\ \times (112)_{10} \\ \hline 670 \\ 3350 \\ 33500 \\ \hline (37520)_{10} \end{array}$$

$$(447)_{10}$$

$$= (546)_9$$

$$\begin{array}{r} 9 \overline{) 447} \\ 9 \overline{) 49} \quad \text{---} 6 \quad \uparrow \\ 9 \overline{) 5} \quad \text{---} 4 \\ 0 \quad \text{---} 5 \end{array}$$

$$(223)_{10}$$

$$= (267)_9$$

$$\begin{array}{r} 9 \overline{) 223} \\ 9 \overline{) 24} \quad \text{---} 7 \quad \uparrow \\ 9 \overline{) 2} \quad \text{---} 6 \\ 0 \quad \text{---} 2 \end{array}$$

$$(37520)_{10}$$

$$= (56418)_9$$

$$\begin{array}{r}
 9 \overline{) 37520} \\
 \underline{9 4168} 8 \\
 9 \overline{) 163} 1 \\
 \underline{9 51} 4 \\
 9 \overline{) 5} 6 \\
 \underline{0} 5
 \end{array}$$

Since, Addition, subtraction and multiplication for the following base -9 numbers are verified after converting the problem into decimal.

[verified]

Answer to the question no 4

a)

$$(379)_{10} \longrightarrow (101111011)_2$$

$$= (000\ 0001\ 0111\ 1011)_2$$

$$2 \overline{) 379}$$

$$2 \overline{) 189} - 1$$

$$2 \overline{) 94} - 1$$

$$2 \overline{) 47} - 0$$

$$2 \overline{) 23} - 1$$

$$2 \overline{) 11} - 1$$

$$2 \overline{) 5} - 1$$

$$2 \overline{) 2} - 1$$

$$2 \overline{) 1} - 0$$

$$0 - 1$$

$$= (000\ 0001\ 0111\ 1011)_{16}$$

$$= (000\ 0001\ 0111\ 1011)_{25}$$

$$(98)_{10} \longrightarrow (1100010)_2$$

$$2 \overline{) 98}$$

$$2 \overline{) 49} - 0$$

$$2 \overline{) 24} - 1$$

$$2 \overline{) 12} - 0$$

$$2 \overline{) 6} - 0$$

$$2 \overline{) 3} - 0$$

$$2 \overline{) 1} - 1$$

$$0 - 1$$

$$= (000\ 0000\ 0110\ 0010)_2$$

$$= (000\ 0000\ 0110\ 0010)_{16}$$

$$= (000\ 0000\ 0110\ 0010)_{25}$$

one's complement system:

$$\begin{array}{rcccc} & 000 & 0001 & 0111 & 1011 \\ + & 000 & 0000 & 0110 & 0010 \\ \hline & 000 & 0001 & 1101 & 1101 \end{array}$$

two's complement system:

$$\begin{array}{rcccc} & 000 & 0001 & 0111 & 1011 \\ + & 000 & 0000 & 0110 & 0010 \\ \hline & 000 & 0001 & 1101 & 1101 \end{array}$$

So,

No overflow in both cases.

Answer to the question no 5

$$\text{Each RAM costs} = (102)_{16}$$

$$= (1 \times 16^2) + (12 \times 16^1) + (2 \times 16^0)$$

$$= (450)_{10}$$

$$\begin{array}{r} (450)_{10} \\ \times (2)_{10} \\ \hline (900)_{10} \end{array}$$

$$\therefore \text{Two RAM costs} = (900)_{10}$$

$$\text{A graphics card costs} = (10010110000)_2$$

$$= 1 \times 2^{10} + 0 \times 2^9 + 0 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 \\ + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= (1200)_{10}$$

The amount of money my friend gave me

$$= (4064)_8$$

$$= 4 \times 8^3 + 0 \times 8^2 + 6 \times 8^1 + 4 \times 8^0$$

$$= (2100)_{10}$$

Total expense:

$$\begin{array}{r} (1200)_{10} \\ + (900)_{10} \\ \hline (2100)_{10} \end{array}$$

$$\text{Money left} = (2100)_{10} - (2100)_{10} \\ = (0)_{10}$$

so, I have left nothing /
 $(0)_{10}$ dollars after buying those
components.